

CROSSSAFE · REQUIREMENTS

CrossSafe requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

CrossSafe requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users or a road authority, and will be checked by calculation at TRL 3 and revised after co-design (see CRS-PRB-001). Where a local rule is stricter, the local rule wins.

The **design case** is a marked crossing on a two-lane road 7 m kerb to kerb, one CrossSafe assembly on each side, 300 activations of 20 s per day, 2.5 peak sun hours per day in the worst month, and an ambient range of -20 to +50 °C.

Table 1. Requirements and status against the TRL 2 estimates in CRS-PRC-001.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Warning light geometry and pattern	Two amber rectangular indications per face, each at least 127 x 51 mm (5 x 2 in); wig-wag pattern at 75 flashing sequences per minute, as in FHWA IA-21; pattern set in firmware for other jurisdictions	Design review against the pilot jurisdiction's rules	Met by design
R2	Visibility	Indications distinguishable by a driver at 100 m or more in direct sun and not glaring at night (automatic dimming)	Photometric calculation from LED data; later field check	Unverified
R3	Activation	Push button and passive presence detection; both sides of the road flash within 0.5 s of a trigger on either side	Timing budget; later bench test	Met by estimate (radio link about 0.1 s)
R4	Flash duration	Set per site from crossing length / 0.9 m/s + 5 s, adjustable 10 to 40 s; a new trigger while flashing extends the time	Firmware review	Met by design (about 13 s for 7 m)
R5	Detection quality	Passive detection triggers for a person waiting 1 s or more in the waiting zone; false activations 5 % or fewer of all activations; missed waiting pedestrians 2 % or fewer	Site trial with manual counts	Unverified
R6	Autonomy without sun	5 days or more of design-case use from a full battery	Energy calculation	Not met: about 4.1 days (estimate)
R7	Energy balance	Energy neutral at 2.5 peak sun hours; recover from 20 % to full in 3 days or fewer at 5 peak sun hours	Solar yield calculation	Met by estimate (break-even about 2.0 h; recovery about 2.7 days)
R8	Environment	Enclosures and connectors IP65 or better; operate -20 to +50 °C ambient; battery charges only at 0 to 45 °C cell temperature	Datasheets; later spray and thermal tests	Met by design, thermal margin in sun unverified
R9	Structure	Post and clamps carry the sign and panel in a 40 m/s gust with stress 60 % or less of yield	Wind load calculation	Not met with the modeled 89 x 4 mm post: about 169 MPa, 72 % of S235 yield (estimate); met with a 114.3 x 3.6 mm post (about 110 MPa, 47 %)
R10	Mounting	Fits the project's own post or an existing 60 to 114 mm pole with band clamps; sign bottom at 2.1 m or more (or local rule); two-person crew installs one assembly in 2 h or less	Design review; later timed trial	Unverified (install time)
R11	Accessibility	Push button 0.9 to 1.1 m above the walking surface, operable with 22 N or less, with an audible and tactile acknowledgement and a pedestrian-facing confirmation light	Design review with disability groups	Met by design, to be checked in co-design
R12				Met by design

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
	Fail-safe behavior	Never flashes continuously; a fault (low battery, open LED, lost radio link) is shown on a maintenance indicator and logged; with the link lost, each side still flashes on a local trigger	Firmware review; fault injection at TRL 4	
R13	Privacy	No images or audio are recorded or leave the device; only activation counts, faults and battery state are logged or sent	Design review	Met by design
R14	Theft and vandal resistance	Battery, charger and controller 2.8 m or more above the walking surface; security fasteners; no exposed cables below 2.5 m	Design review	Met by design (enclosure at about 2.85 m)
R15	Affordable	Parts for one assembly \$350 or less; parts for a two-sided crossing \$700 or less	Priced BOM (bom/ bom.csv)	Not met: about \$361 per assembly, about \$722 per crossing (estimate); about \$311 per assembly on an existing pole

■ Requirements not met or at risk

- **R6 not met.** About 123 Wh usable against about 30 Wh/day gives about 4.1 days. Options are in CRS-PRC-001 (gate the radar with a PIR sensor, or fit an 18 Ah battery).
- **R15 not met.** About 3 % over for one assembly and for a crossing when the project supplies its own posts.
- **R9 not met with the modeled post.** A 76 x 3.2 mm post would yield (about 287 MPa); the modeled 89 x 4 mm post reaches about 72 % of yield against a 60 % target. A 114.3 x 3.6 mm post, a 600 mm sign, or mounting on an existing lighting pole meets it; see CRS-PRC-001.
- **R2 and R5 are unverified** and carry the most risk: they decide whether drivers respond and whether passive detection is trusted.

■ Assumptions

- LED heads draw about 6 W each when lit in daytime and are lit about 35 % of each flash cycle; LED drivers are about 91 % efficient.
- Standby draw about 0.6 W: radar about 0.4 W, controller and receiving radio about 0.15 W, charger about 0.05 W.
- Solar system efficiency about 76 % from panel rating to energy available to store (temperature, dust, wiring, MPPT and charging).
- Walking speed of 0.9 m/s allows for children and older people; the US MUTCD basis is faster, and local rules decide.
- Gust wind speed of 40 m/s and a drag coefficient of 1.2 for flat plates; S235 steel for the post.